

Notes – Precedence in Expression Evaluation

Name _____

For each expression below,

- Parenthesize if necessary,
- Label each in operand and operator in order of evaluation
- Build an expression tree
- Evaluate expression or give final values of variables as indicated

Note: Assume undefined methods and variables are appropriately defined.

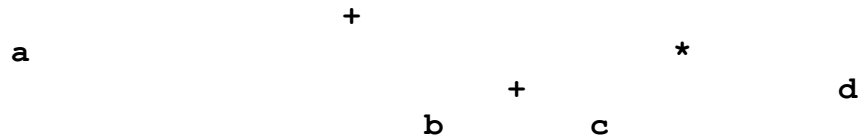
Use int a=2, b=4, c=5, d=3;

1. `a + (b + c) * d`

Evaluation order:

a	+	(b	+	c)	*	d
1	7	2	4	3	6	5

Expression Tree:



Value: 29

Use int a=2, b=4, c=5, d=3;

2. `c + a * Math.round(2.6 + b)`

Use int a=2, b=4, c=5, d=3;

3. `(a + b) + ++b`

Use int a=2, b=4, c=5, d=3;

4. `c /= ++c;` ← This actually includes the type cast shown below:

`c = (int)(c / ++c)`

Use int a=2, b=4, c=5, d=3;

5. $(b = 2 * b) + (b + b)$

Use int a=2, b=4, c=5, d=3;

6. $\text{int } f = a > 0 ? (\text{int})2.7 : (\text{int})3.7;$

Use int a=2, b=4, c=5, d=3;

7. $\text{int}[] \text{ g} = \{6, 4, 2, 9, 3, 2, 0, 6, 8, 9, 5\};$
 $\text{out.println}(\text{g}[\text{g}[b] + \text{g}[a]]);$

Use int a=2, b=4, c=5, d=3;

8. $\text{int}[] \text{ g} = \{6, 4, 2, 9, 3, 2, 0, 6, 8, 9, 5\};$
 $\text{int}[] \text{ h} = \{23, 49, 1, 14, 93\};$
 $\text{out.println}(\text{g}[\text{g}[b] + (\text{g}=\text{h})[a]]);$